

Based on the submission rules in your screenshot, the abstract needs to be **≤350 words** and clearly cover **objectives, methodology, results, and conclusions**. Combining **WP5.1 Process Monitoring + WP5.3 Process Twin/HMI** actually gives you a stronger conference story than presenting WP5.3 alone.

Which category?

I would submit this primarily under **Innovation**, with **Quality** as the secondary theme.

The reason is that the strongest contribution is not simply improving cycle time or resource utilization. It is the architecture that converts heterogeneous AFP process and machine data into a **contextualized process twin** combining live monitoring, spatial analysis, time-series analysis, and engineering visualization.

Quality is also strongly represented because you are looking at process variations, KPI/KPP trends, TCP/joint deviations and anomaly detection. **Efficiency** is relevant—for example, faster engineering investigation and reduced manual analysis—but unless you already have measured efficiency gains, I would not make it the primary category.

A good positioning would therefore be:

Primary: Innovation | Supporting themes: Quality and Efficiency

Proposed title

From Process Data to Engineering Insight: An Integrated Process Monitoring and Digital Twin Framework for Automated Fiber Placement

Alternative, slightly more technical:

A Data-Driven Process Twin Architecture for Spatial and Time-Series Monitoring of Automated Fiber Placement

I prefer the first one for TechXceed because it communicates the engineering value rather than sounding like a purely software-architecture paper.

Proposed abstract

From Process Data to Engineering Insight: An Integrated Process Monitoring and Digital Twin Framework for Automated Fiber Placement

Automated Fiber Placement (AFP) generates large volumes of heterogeneous process, machine and trajectory data during manufacturing. Although these data provide valuable information about process behaviour, transforming them into contextualized engineering insight

requires synchronized acquisition, structured data management, analytical methods and intuitive visualization. This work presents an integrated **Process Monitoring and Process Twin framework** that combines these capabilities to support data-driven understanding of AFP operations.

The proposed framework integrates process-data acquisition, synchronization, filtering and storage with a data-driven Process Twin and Human-Machine Interface (HMI). Process parameters and machine information are associated with manufacturing context such as product, part, layer, course, strip and tow, enabling analysis at different levels of process granularity. The Process Twin provides three complementary analytical perspectives. First, programmed and actual TCP and joint trajectories are synchronized to quantify motion deviation and spatially map variations onto the 3D part geometry. Second, process KPIs such as tow tension, pressure, temperature and process speed are analysed and associated with their corresponding manufacturing locations, enabling KPI trends and variations to be visualized directly on the product geometry. Third, time-series analysis is applied to process signals to identify abnormal temporal behaviour such as excursions, spikes and drift.

The architecture additionally incorporates machine-health monitoring, maintenance-oriented Pareto analysis and a framework for predictive-maintenance information. A modular RXU-based software architecture separates machine connectivity, data management, process analytics and visualization while supporting both continuous data streams and query-based historical analysis.

The resulting framework establishes a common digital thread from AFP process data to temporal and spatial engineering insight. By combining process monitoring with contextualized 3D visualization and data-driven analysis, the approach aims to improve process transparency, accelerate investigation of manufacturing variations and provide a scalable foundation for future AFP process optimization.

Word count: ~275 words

One important adjustment before submission

Because your screenshot specifically says the abstract should include **results**, I would be careful with wording such as “*The resulting framework establishes...*” if the complete WP5.1/WP5.3 implementation has not yet been validated.

If you already have prototype results—such as successful ingestion of sample mTorres/AFP data, TCP deviation demonstrated on 3D geometry, KPI visualization, or anomaly detection on sample time-series data—we should add **2–3 concrete results**. That will make the abstract considerably stronger and more credible for selection among the top eight.

For example, instead of only saying “*The resulting framework establishes...*”, we could write something like:

“Initial prototype validation demonstrates synchronized visualization of programmed and actual TCP trajectories with spatial deviation mapping on 3D geometry, together with contextualized KPI and time-series analysis using representative AFP datasets.”

But I would include that only if you have actually demonstrated those elements.

I can also prepare **three competing ≤350-word abstracts—Innovation-focused, Quality-focused, and Efficiency-focused**—so you can choose the version most likely to be selected for TechXceed.

I would like to put in quality

Because it decides the quality of the layup

Identify process anomalies and identify the defects such as missing tow, wrinkles, gaps etc

For your reference attaching the overview of next gen layup project also

Quality is the stronger category with this framing.

The NextGen Laying overview strengthens that choice: it explicitly positions AFP inline monitoring, digitalisation/sensing, robustness, productivity, and reduced manual intervention as part of the wider project. For your WP5.1 + WP5.3 contribution, the distinctive quality story is: **capture what happened during layup → contextualize it → identify process variation/anomalies → relate it spatially to the manufactured part → support quality assessment and investigation.**

One caution: I would not claim that your WP5.3 time-series model itself “detects wrinkles, gaps, missing tow” unless those defect-detection capabilities are actually implemented. A stronger and technically accurate formulation is that process anomalies and inspection/quality observations—including missing tow, gaps or wrinkles where available—can be correlated and spatially contextualized to support quality assessment.

Category: Quality

Suggested title

From Process Monitoring to Quality Insight: A Data-Driven Process Twin for Automated Fiber Placement

I think this is stronger than putting “Digital Twin” alone in the title because it immediately connects your work to the conference category.

Abstract

Automated Fiber Placement (AFP) is a key technology for manufacturing large composite aerostructures, where consistent layup quality depends on maintaining process and machine behaviour within the intended operating conditions. Variations in parameters such as tow tension, compaction pressure, temperature, laying speed and machine trajectory can influence layup quality and may be associated with manufacturing conditions requiring further investigation. As AFP systems increase in productivity and automation, efficiently transforming large volumes of process data into actionable quality information becomes increasingly important.

This work presents an integrated **Process Monitoring and data-driven Process Twin framework** for quality-oriented monitoring of AFP manufacturing. The approach combines synchronized acquisition and storage of machine and process data with contextualized process analysis and visualization. Manufacturing data are associated with product, part, layer, course, strip and tow information, enabling quality-relevant behaviour to be investigated at different levels of the layup process.

The framework provides complementary spatial and temporal quality perspectives. Programmed and actual TCP and joint trajectories are compared to quantify execution deviations and spatially visualize their variation on the 3D part geometry. Process KPIs, including tow tension, pressure, temperature and laying speed, are analysed and mapped to their corresponding locations on the manufactured geometry. Time-series analysis is used to identify process variations such as excursions, spikes and drift. These results can be combined with available quality observations, including conditions such as missing tows, gaps or wrinkles, to support investigation of relationships between process behaviour and observed layup quality.

A modular RXU-based architecture integrates connectivity, data management, analytics and HMI visualization while supporting both continuous process monitoring and historical analysis. The proposed framework establishes a digital thread between **manufacturing execution, process variation and spatial quality context**, supporting earlier identification and investigation of quality-relevant conditions and providing a foundation for data-driven improvement of next-generation AFP manufacturing.

~290 words

Why I prefer this version for Quality

The story now has a clear progression:

AFP layup → process/machine data → KPI variation → spatial + temporal analysis → quality context → engineering investigation

That is stronger for **Quality** than describing WP5.1/WP5.3 primarily as a software or HMI architecture.

I would also avoid making **predictive maintenance, Pareto, Grafana, QuestDB, protobuf, etc.** prominent in this abstract. They matter for your architecture, but they dilute the conference message. For this submission, the compelling contribution is **how WP5.1 + WP5.3 use manufacturing data to understand and protect AFP layup quality.**

One phrase from the NextGen Laying overview is especially useful for positioning: the project calls out “**AFP/ATL inline monitoring**” alongside digitalisation and sensing. Your work can be presented as the next layer that turns that monitored information into **quality-oriented engineering insight.**

I would not claim that your WP5.3 time-series model itself “detects wrinkles, gaps, missing tow” unless those defect-detection capabilities are actually implemented. A stronger and technically accurate formulation is that process anomalies and inspection/quality observations—including missing tow, gaps or wrinkles where available—can be correlated and spatially contextualized to support quality assessment.

This is in plan as part of online inspection
Along with this time series and root cause analysis

Yes. That changes the framing significantly. If **online inspection for defects such as missing tow, gaps and wrinkles is explicitly planned**, alongside **time-series process anomaly detection and root-cause analysis**, then you can legitimately position the combined WP5.1 + WP5.3 work much more strongly under **Quality**.

The important thing is to distinguish the three levels rather than implying that time-series analysis itself detects physical defects:

Online inspection → physical layup defect detection
Time-series analysis → process anomaly detection
Root-cause analysis → relationship/investigation between process behaviour and detected quality defects

That actually gives you a stronger conference story:

Detect the defect → identify what process anomaly occurred → investigate why it occurred → spatially correlate everything on the manufactured part.

I would revise the abstract accordingly

From Process Monitoring to Quality Insight: An Integrated Data-Driven Framework for Automated Fiber Placement

Automated Fiber Placement (AFP) is a key technology for manufacturing large composite aerostructures, where consistent layup quality depends on both accurate material deposition and stable process behaviour. As AFP systems evolve towards higher productivity and increased automation, early identification of layup defects and process variations becomes essential for improving manufacturing quality while reducing dependence on manual inspection and investigation.

This work presents an integrated **quality-oriented Process Monitoring and data-driven Process Twin framework** for next-generation AFP manufacturing. The approach combines synchronized acquisition and management of machine, process and inspection data with online quality monitoring, process analytics and contextualized visualization. Manufacturing information is associated with product, part, layer, course, strip and tow context, enabling quality events and process behaviour to be traced to their corresponding manufacturing location.

The framework addresses quality from complementary perspectives. **Online inspection** is used to identify layup quality conditions such as missing tows, gaps and wrinkles. In parallel, **time-series analysis** monitors process parameters such as tow tension, compaction pressure, temperature and laying speed to identify abnormal behaviour including excursions, spikes and drift. Programmed and actual TCP and joint trajectories are compared to quantify execution deviations, while process KPI variations and trajectory deviations are spatially mapped onto the 3D part geometry.

By correlating detected defects with process anomalies, machine behaviour and manufacturing context, the framework further supports **root-cause investigation**, enabling engineers to explore which process conditions may have contributed to an observed quality issue rather than treating defect detection as an isolated activity.

A modular RXU-based architecture integrates machine connectivity, data management, online monitoring, analytics and visualization, supporting both continuous monitoring and historical investigation. The resulting digital thread connects **process execution, anomaly detection, online inspection and engineering investigation**, providing a foundation for earlier quality assessment and systematic improvement of AFP layup processes.

~300 words

I think this is a **much stronger Quality submission** because now the contribution isn't merely "we monitor KPIs." It presents a coherent quality chain:

Process execution → Online inspection → Defect detection → Process anomaly detection → Spatial correlation → Root-cause investigation → Quality insight

And I would retain "**root-cause investigation**" rather than "automatic root-cause identification." That remains consistent with your WP5.3 scope: the system provides the data correlations and engineering framework to support an engineer's investigation rather than claiming autonomous diagnosis.

Yes. The NextGen Laying reference slide contains a few **strong quantitative figures**, and using one or two would strengthen the abstract—especially the motivation/results context.

From the slide, the clearest numbers are:

- **>100 layup machines** potentially required for all CFRP components under the e-Action context if productivity is not sufficiently improved.

- **>1 m/s axis speed** is mentioned as the level at which existing machines are already reaching maximum performance in terms of pure axis speed.
- Material formats are indicated as approximately **¼" to 2"**.
- The stated project benefit target is **20% more productivity**.
- The slide also states a target of **>50% RC reduction**, attributed to lead-time reduction and reduction in the number of operators.

For your **Quality** abstract, however, I would not insert all these numbers. The **20% productivity** and **>50% RC reduction** are NextGen Laying **overall project benefit targets**, not demonstrated results of WP5.1/WP5.3. We should avoid wording that implies your Process Twin has already achieved them.

A strong way to use the numbers is in the **motivation**:

“Next-generation AFP manufacturing targets a 20% productivity improvement, while the projected industrial scale may require more than 100 layup machines across CFRP components. At this scale, maintaining consistent layup quality through predominantly manual monitoring and investigation becomes increasingly challenging.”

Then transition directly into your quality contribution:

“This motivates an integrated approach combining online inspection, process monitoring and data-driven analysis to identify quality defects and abnormal process behaviour at the point of manufacture.”

I particularly like the **>100 machines** number for your abstract because it explains *why scalable automated quality monitoring matters*. The 20% figure then explains the productivity pressure under which quality still has to be maintained.

Revised opening using the NextGen numbers

I would change the first paragraph of our abstract to:

Automated Fiber Placement (AFP) is a key technology for manufacturing large composite aerostructures, where increasing production rates must be achieved without compromising layup quality. Within the NextGen Laying initiative, a 20% productivity improvement is targeted, while the projected industrial scale may require more than 100 layup machines across CFRP components. At this scale, consistent quality assurance cannot rely solely on conventional manual monitoring and post-process investigation. Early identification of layup defects, process variations and their potential relationships therefore becomes increasingly important.

Then the rest of the abstract can introduce:

Online inspection → missing tow, gaps, wrinkles
Process monitoring → tension, pressure, temperature, speed
Time-series analysis → excursions, spikes, drift
TCP/joint deviation → spatial execution variation
3D Process Twin → spatial contextualization
Root-cause investigation → defect ↔ process behaviour correlation

That gives you a much more compelling **Quality** narrative than simply stating that AFP generates a lot of data.

One caution: I'd phrase both **20%** and **>100 machines** as *NextGen Laying targets/projections*, not as results achieved by your work. The slide supports the former, not the latter.

Yes. Since you want to submit under **Quality**, I would make the abstract explicitly connect the NextGen Laying scale/productivity motivation to the quality chain: **online inspection** → **process anomaly detection** → **spatial correlation** → **root-cause investigation**.

I would also make sure the **20% productivity improvement** and **>100 layup machines** are presented as NextGen Laying targets/projections, not as achievements of WP5.1/WP5.3.

Proposed title

From Process Monitoring to Quality Insight: An Integrated Data-Driven Framework for Next-Generation Automated Fiber Placement

Abstract — Quality Category

Automated Fiber Placement (AFP) is a key technology for manufacturing large composite aerostructures, where increasing production rates must be achieved without compromising layup quality. Within the NextGen Laying initiative, a **20% productivity improvement is targeted**, while the projected industrial scale may require **more than 100 layup machines** across CFRP components. At this scale, maintaining consistent manufacturing quality requires scalable methods for detecting layup defects, monitoring process behaviour and supporting engineering investigation.

This work presents an integrated **Process Monitoring and data-driven Process Twin framework** for quality-oriented monitoring of AFP manufacturing. The approach combines synchronized acquisition and management of machine, process and inspection data with online quality monitoring, process analytics and contextualized visualization. Data are associated with manufacturing context—including product, part, layer, course, strip and tow—enabling process behaviour and quality events to be traced to their corresponding manufacturing location.

The framework addresses quality through complementary analytical capabilities. **Online inspection** identifies layup quality conditions such as missing tows, gaps and wrinkles. In parallel, **time-series analysis** monitors process parameters such as tow tension, compaction pressure, temperature and laying speed to identify abnormal behaviour including excursions, spikes and drift. Programmed and actual TCP and joint trajectories are compared to quantify execution deviations. Process KPI variations, trajectory deviations and detected quality conditions are spatially contextualized on the **3D part geometry**, providing a unified view of manufacturing execution and layup quality.

By correlating detected defects with process anomalies, trajectory deviations and manufacturing context, the framework further supports **root-cause investigation**, helping engineers identify process conditions associated with observed quality issues. A modular RXU-based architecture integrates connectivity, data management, monitoring, analytics and visualization while supporting both continuous and historical analysis.

The proposed framework establishes a digital thread connecting **process execution, online inspection, anomaly detection and engineering investigation**, providing a scalable foundation for earlier quality assessment and continuous improvement of next-generation AFP manufacturing.

Approx. 320 words — within the 350-word limit.

I would use this version over the earlier one. It makes **Quality** unmistakably the central contribution while still bringing in the NextGen project's productivity motivation without claiming that WP5.1/WP5.3 themselves achieved the 20% target.

Not explicitly. In the last abstract I focused mainly on **process quality**: online inspection, KPI/time-series anomalies, TCP/joint deviation, 3D spatial visualization, and root-cause investigation.

Given your combined **WP5.1 + WP5.3** scope, I think we should mention **machine health monitoring and predictive maintenance**, but keep them secondary. Otherwise the abstract may become too broad and dilute the **Quality** theme.

A good addition would be in the architecture paragraph:

“The framework additionally integrates machine-health monitoring and predictive-maintenance information, enabling process-quality observations to be considered alongside equipment condition and maintenance indicators.”

Then the overall quality chain becomes:

Machine/process monitoring → Machine health → Predictive maintenance → Process anomaly detection → Online defect detection → Spatial correlation → Root-cause investigation → Quality insight

I would **not emphasize the Pareto dashboard or Grafana** in the abstract; those are implementation/visualization details better suited to the presentation.

Revised paragraph

I would replace the fourth paragraph with:

By correlating detected defects with process anomalies, trajectory deviations and manufacturing context, the framework supports **root-cause investigation**, helping engineers identify process conditions associated with observed quality issues. The framework additionally incorporates **machine-health monitoring and predictive-maintenance information**, enabling process-quality observations to be considered together with equipment condition and maintenance indicators. A modular RXU-based architecture integrates connectivity, data management, online inspection, process and machine-health analytics, and visualization while supporting both continuous monitoring and historical investigation.

And I would slightly strengthen the conclusion:

The proposed framework establishes a digital thread connecting **process execution, machine health, online inspection, anomaly detection and engineering investigation**, providing a scalable foundation for earlier quality assessment, improved process understanding and continuous improvement of next-generation AFP manufacturing.

This way **predictive maintenance and health check are represented**, but the paper remains clearly a **Quality** submission rather than becoming a maintenance-focused abstract.

For the **Quality** category, I would keep the quality chain as the central contribution and include machine health/predictive maintenance as supporting capabilities rather than letting maintenance dominate the abstract.

Proposed Title

From Process Monitoring to Quality Insight: An Integrated Data-Driven Framework for Next-Generation Automated Fiber Placement

Abstract

Automated Fiber Placement (AFP) is a key technology for manufacturing large composite aerostructures, where increasing production rates must be achieved without compromising layup quality. Within the NextGen Laying initiative, a **20% productivity improvement is targeted**, while the projected industrial scale may require **more than 100 layup machines** across CFRP components. At this scale, maintaining consistent manufacturing quality requires scalable methods for detecting layup defects, monitoring process and machine behaviour, and supporting engineering investigation.

This work presents an integrated **Process Monitoring and data-driven Process Twin framework** for quality-oriented monitoring of AFP manufacturing. The approach combines synchronized acquisition and management of machine, process and inspection data with online quality monitoring, process analytics and contextualized visualization. Data are associated with manufacturing context—including product, part, layer, course, strip and tow—enabling process behaviour and quality events to be traced to their corresponding manufacturing location.

The framework addresses quality through complementary analytical capabilities. **Online inspection** identifies layup quality conditions such as missing tows, gaps and wrinkles. In parallel, **time-series analysis** monitors process parameters such as tow tension, compaction pressure, temperature and laying speed to identify abnormal behaviour including excursions, spikes and drift. Programmed and actual TCP and joint trajectories are compared to quantify execution deviations. Process KPI variations, trajectory deviations and detected quality conditions are spatially contextualized on the **3D part geometry**, providing a unified view of manufacturing execution and layup quality.

By correlating detected defects with process anomalies, trajectory deviations and manufacturing context, the framework supports **root-cause investigation** of observed quality issues. The framework additionally incorporates **machine-health monitoring and predictive-maintenance information**, enabling process-quality observations to be considered alongside equipment condition and maintenance indicators. A modular **RXU-based architecture** integrates connectivity, data management, online inspection, process and machine-health analytics, and visualization while supporting both continuous monitoring and historical investigation.

The proposed framework establishes a digital thread connecting **process execution, machine health, online inspection, anomaly detection and engineering investigation**, providing a scalable foundation for earlier quality assessment, improved process understanding and continuous improvement of next-generation AFP manufacturing.

Word count: ~325 words — within the 350-word limit.

I think this is now well balanced for **Quality**: defect detection and layup quality remain the main story, while **process anomaly detection, TCP/joint deviation, root-cause investigation, machine health and predictive maintenance** show that the solution covers quality from both the **process and equipment perspectives**.

